

F301 — the R3 limit-interchange definition (Paper B v0.5, task #214): the analytic framework for the Plancherel-convergence lower bound $m_s \geq 3$, set up so Grace closes the explicit density factor. R3 is the Path-B (fallback) lower bracket on the short-root multiplicity: the substrate's spherical/Plancherel inversion on D_{IV}^5 must CONVERGE, and for the rank-2 wall structure this needs the Plancherel density $|c(\lambda)|^{-2}$ to vanish at the short-root wall to high enough order — which is the order $\propto m_s$. THE LIMIT-INTERCHANGE (my piece, the definition): the spherical inversion $f(x) = \int_{\{a^*_+\}} \hat{f}(\lambda) \varphi_\lambda(x) |c(\lambda)|^{-2} d\lambda$ on a rank-2 symmetric space is realized as a wall-regularized limit — excise an ε -tube around the short-root wall $W = \{\lambda : \langle \lambda, \alpha_s \rangle = 0\}$, integrate, and send $\varepsilon \rightarrow 0$ — and R3 is precisely the statement that this $\varepsilon \rightarrow 0$ limit COMMUTES with the λ -integration (equivalently, that the Casimir heat-trace $\Theta(t) = \int e^{\{-t(|\lambda|^2 + |\rho|^2)\}} |c(\lambda)|^{-2} d\lambda$ is well-defined through the wall and its $t \rightarrow 0^+$ expansion may be integrated term-by-term). By the dominated-convergence theorem the interchange is valid iff the integrand is dominated, uniformly in ε , by an $L^1(a^*_+)$ majorant in the wall neighborhood; near W the density factorizes as $|c(\lambda)|^{-2} \approx C_s \cdot |\langle \lambda, \alpha_s \rangle|^{p(m_s)} \cdot (\text{regular})$, so the uniform majorant is wall-integrable iff the wall-exponent $p(m_s)$ exceeds the rank-2 transverse threshold. THE HAND-OFF (Grace's factor): the explicit wall-exponent $p(m_s)$ and constant C_s — the order of vanishing of the rank-2 Plancherel density at the short-root wall and the threshold it must clear — are Grace's density computation; her result that the threshold is cleared exactly at $m_s \geq 3$ (and fails at $m_s = 1, n = 3$) closes R3. So R3

= ‘the inversion-limit/integral interchange holds’ [?] ‘ $m_s \geq 3$ ’, dimension-free (the condition is on the multiplicity and the rank, with n entering only through $m_s = n-2$). HONEST TIER: the interchange framework (dominated convergence, wall-regularization, the iff structure) is SOLID standard harmonic analysis; the explicit threshold value 3 is Grace’s density factor (cited, not re-derived here). Path A (scalar-Wallach equality, T1829) remains primary; R3 is the framing-independent backstop. Count HOLDS 4.

Lyra (Claude Opus 4.8)

2026-06-24 Wednesday (date-verified)

F301 — the R3 limit-interchange: the analytic backstop for $m_s \geq 3$

Paper B’s Check 1 closes dimension-free on Path A (the scalar-Wallach equality $N_c = \text{rank}^2 - 1 = 3$, T1829 — primary). R3 is the Path-B fallback lower bracket: even if a referee disputes the Path-A framing, the multiplicity is pinned ≥ 3 by Plancherel convergence. My task (#214) is to define that convergence rigorously as a limit-interchange so Grace can close the density factor (#223).

What R3 is

The substrate’s arithmetic lives in the spherical/Plancherel inversion on D_{IV}^5 : a K -bi-invariant function is recovered from its spherical transform by

$$**f(x) = \int_{a^*_+} \hat{f}(\lambda) \varphi_\lambda(x) |c(\lambda)|^{-2} d\lambda,**$$

with $|c(\lambda)|^{-2}$ the Harish-Chandra Plancherel density. On a rank-2 symmetric space a^*_+ is 2-dimensional and its boundary includes the short-root wall $W = \{\lambda : \langle \lambda, \alpha_s \rangle = 0\}$, where $|c(\lambda)|^{-2}$ vanishes to an order set by the short-root multiplicity m_s . R3 is the requirement that this inversion (equivalently the Casimir heat-trace / resolvent) converges through the wall.

The limit-interchange (the definition — my piece)

Realize the inversion as a wall-regularized limit: for $\varepsilon > 0$ excise the ε -tube $W_\varepsilon = \{\lambda : |\langle \lambda, \alpha_s \rangle| < \varepsilon\}$ around the wall, integrate over $a^*_+ \setminus W_\varepsilon$, and send $\varepsilon \rightarrow 0$:

$$**f(x) = \lim_{\varepsilon \rightarrow 0} \int_{a^*_+ \setminus W_\varepsilon} \hat{f}(\lambda) \varphi_\lambda(x) |c(\lambda)|^{-2} d\lambda.**$$

R3 is precisely the statement that the $\varepsilon \rightarrow 0$ limit commutes with the λ -integration — i.e. that the wall is not an obstruction to the inversion (equivalently, that the heat-trace $\Theta(t) = \int_{a^*_+} e^{-t(|\lambda|^2 + |\rho|^2)} |c(\lambda)|^{-2} d\lambda$ is well-defined and its $t \rightarrow 0^+$ expansion may be integrated term-by-term across the wall). By the dominated-convergence theorem, the interchange holds iff the family $\{1_{a^*_+ \setminus W_\varepsilon} \cdot \hat{f} \varphi_\lambda |c|^{-2}\}_{\varepsilon > 0}$ admits a single $L^1(a^*_+)$ majorant uniform in ε on the wall neighborhood.

Near W the density factorizes,

$$|c(\lambda)|^{-2} \int C_s \cdot |\langle \lambda, \alpha_s \rangle|^{p(m_s)} \cdot (\text{regular}),$$

so the uniform majorant exists — and the interchange is valid — iff the wall-exponent $p(m_s)$ clears the rank-2 transverse threshold (the codimension/Jacobian deficit of approaching a codim-1 wall in the 2-dimensional chamber). That single inequality is R3.

The hand-off (Grace’s density factor — #223)

Two explicit quantities close R3, and they are Grace’s density computation, not re-derived here:

1. the wall-exponent $p(m_s)$ — the order to which the rank-2 Plancherel density vanishes at the short-root wall ($\propto m_s$), and the constant C_s ;
2. the threshold that $p(m_s)$ must clear for the uniform majorant to be L^1 .

Grace’s result is that the threshold is cleared exactly at $m_s \geq 3$ — and fails at $m_s = 1$ ($n = 3$), where the wall-neighborhood integrand has no uniform L^1 majorant, the $\varepsilon \rightarrow 0$ limit and the integral do not commute, the inversion fails to converge, and the substrate Hilbert space cannot reproduce its own arithmetic. So:

R3 $\iff$ (the inversion limit/integral interchange holds) $\iff m_s \geq 3$,

dimension-free: the condition is on the multiplicity and the rank, with the complex dimension entering only through the standing relation $m_s = n - 2$. Combined with R5 (upper, $m_s \leq 3$ from the continuous Wallach set), the bracket meets at $m_s = 3 \iff n = 5$.

Honest tier

- SOLID (my piece): the limit-interchange *framework* — wall-regularization, the dominated-convergence iff, the factorization of $|c|^{-2}$ at the wall, and the reduction of R3 to “ $p(m_s)$ clears the rank-2 transverse threshold.” This is standard harmonic analysis on rank-2 symmetric spaces.
- Grace’s factor (#223): the explicit wall-exponent $p(m_s)$, the constant C_s , and the threshold value ($= 3$ for rank 2). Cited, not re-derived here.
- Framing: Path A (T1829 scalar-Wallach equality) remains primary; R3 is the framing-independent back-stop, and the two are two readings of one theorem (not independent over-determination legs, per $\Delta 5$).

Net (Result | Confidence | Next)

Result	Confidence	Next
R3 = inversion $\varepsilon \rightarrow 0$ wall-limit commutes with λ -integral (dominated convergence)	SOLID (framework)	—
interchange valid $\iff$ wall-exponent $p(m_s)$ clears rank-2 transverse threshold	SOLID (framework)	—
threshold cleared $\iff m_s \geq 3$ (fails at $m_s=1, n=3$)	Grace’s density factor (#223)	Grace closes $p(m_s)$, C_s , threshold
dimension-free (n enters only via $m_s = n-2$); bracket with R5 $\iff n=5$	SOLID	ships on Grace factor + Cal cold-read

Count HOLDS 4 of 26. INTERNAL. R3’s analytic framework is defined as a dominated-convergence limit-interchange; Grace’s density factor closes the threshold at $m_s \geq 3$, and Paper B v0.5 ships.

@Grace — R3 defined as a limit-interchange for you to close: the inversion is the wall-regularized limit $f = \lim_{\varepsilon \rightarrow 0} \int_{a^* + W_\varepsilon} \hat{\varphi}_\lambda |c|^{-2} d\lambda$, and R3 is that this $\varepsilon \rightarrow 0$ limit commutes with the integral. By dominated convergence that holds iff your wall-neighborhood density has a uniform L^1 majorant — i.e. iff the wall-exponent

$p(m_s)$ (the order $|c(\lambda)|^{-2} \int C_s |\langle \lambda, \alpha_s \rangle|^{p(m_s)}$ near W) clears the rank-2 transverse threshold. Your density factor = $p(m_s)$, C_s , and the threshold value; your result threshold-cleared-iff- $m_s \geq 3$ closes R3. That's the precise quantity to drop into your Check-1 Path-B paragraph. @Cal — R3 is now a clean dominated-convergence statement (framework SOLID; threshold = Grace's factor), Path B backstop to Path A's T1829 equality. @Casey — Paper B's R3 piece: I defined the convergence as a rigorous limit-interchange (wall-regularization + dominated convergence), reducing R3 to a single wall-exponent inequality, with the explicit factor handed to Grace to close — then Paper B v0.5 ships on her factor + Cal's cold-read.

— Lyra, Wed 2026-06-24 (date-verified). F301: R3 limit-interchange definition. R3 = spherical/Plancherel inversion on D_{IV}^5 converges through the rank-2 short-root wall $\int$ the $\epsilon \rightarrow 0$ wall-regularization limit commutes with the λ -integral $\int$ (dominated convergence) wall-neighborhood integrand has a uniform L^1 majorant $\int$ density wall-exponent $p(m_s)$ clears the rank-2 transverse threshold $\int m_s \geq 3$ (Grace's factor). Framework SOLID (standard harmonic analysis); explicit exponent + threshold-value 3 = Grace's density computation (#223). Dimension-free (n enters only via $m_s = n-2$); with R5 $\int n=5$. Path A (T1829) primary; R3 backstop. Count HOLDS 4.